BEING A UNIVERSAL HOMEOMORPHISM IS FPQC LOCAL ON THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file descent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof opens with the author's own 'This can be proved in exactly the same manner as Lemma ...'; the sentence is reproduced verbatim, and the author then supplies the explicit alternative derivation that the counted span carries, so the argument does not rest on that sentence alone.

Editorial note. The counted spans carry 4 navigation stubs; no bibliography citation occurs in the spans.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- fpqc locality on the base for surjectivity, universal injectivity and universal openness. Same chapter, Lemmas lemma-descending-property-surjective, lemma-descending-property-universally-injective and lemma-descending-property-universally-open, invoked at proof lines 5947-5950. Supplies the three locality statements whose conjunction gives the conclusion.
- A map of topological spaces is a homeomorphism exactly when it is injective, surjective and open. Standard topology, used at proof lines 5942-5946. Supplies the characterisation that splits the property into its three components.

Source reference. Original lemma 02KV: sources/stacks-descent/pinned-original-source.tex, source lines 5826–5826. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KS: sources/stacks-descent/pinned-original-source.tex, source lines 5732–5732. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 02KW: sources/stacks-descent/pinned-original-source.tex, source lines 5884–5884. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:descent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 02KT: sources/stacks-descent/pinned-original-source.tex, source lines 5773–5773. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *The property $\mathcal{P}(f) =$"f is a universal homeomorphism" is fpqc local on the base.*

Proof. This can be proved in exactly the same manner as Lemma 02KS. Alternatively, one can use that a map of topological spaces is a homeomorphism if and only if it is injective, surjective, and open. Thus a universal homeomorphism is the same thing as a surjective, universally injective, and universally open morphism. Thus the lemma follows from Lemmas 02KV, 02KW, and 02KT. □